

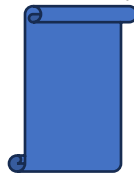


PROJECT OVERVIEW

The **BMW Sales Analysis Dashboard** is an interactive **Power BI business intelligence project** designed to transform raw BMW sales data into meaningful insights about **revenue, unit sales, regional performance, model-level trends and vehicle characteristics**.

The dashboard provides both an **executive-level overview** and a **detailed model-level analysis**, allowing users to move from high-level business performance to specific insights for individual BMW models. Interactive filters and drill-through functionality make it possible to explore how **models, regions, fuel types, transmissions, colors, pricing, mileage, and sales classifications** influence overall sales performance.

The project focuses not only on presenting numbers but also on creating a **clear analytical story** - helping a viewer quickly understand where BMW sales are strongest, which models perform well, how preferences differ across markets, and how sales have evolved over time.



THE PROBLEM STATEMENT:

BMW sales data contains multiple dimensions that can make it difficult to identify important patterns through raw tables alone. The primary challenge was to convert these records into a **business-friendly analytical dashboard** that could answer questions such as:

- Which regions generate the highest revenue and sales volume?
- Which BMW models are performing strongest across different markets?
- Which fuel types and transmission types dominate sales?
- How have revenue and unit sales changed over time?
- Which model - region combinations are performing above or below expectations?
- How do vehicle characteristics such as price, mileage, and engine size vary across models?
- The objective was therefore to build a dashboard that supports **quick performance monitoring, deeper exploration and data-driven decision-making**.
- Where are the major opportunities for **improving revenue, profitability and sales-channel performance**?



DATASET DETAILS

The analysis uses the BMW Sales Dataset sourced from Kaggle. **The dataset contains 5,000 records and 11 fields in the working dataset**, covering important vehicle and sales attributes.

The sales classification field was created to describe “High” and “low” sales and was examined in SQL by creating a “CASE” structure to assign classifications according to sales volume.

Dataset : <https://www.kaggle.com/datasets/yOussefkandil/bmw-sales2010-2024>

The Key fields include:

- Model — BMW vehicle model
- Year — Year of sale
- Region — Geographic sales region
- Color — Vehicle color
- Fuel Type — Petrol, Diesel, Hybrid, Electric, etc.
- Transmission — Vehicle transmission type
- Engine Size (L) — Engine capacity
- Mileage (KM) — Vehicle mileage
- Price (USD) — Vehicle selling price
- Sales Volume — Number of units sold
- Sales Classification — Sales performance classification

The dataset provided a useful combination of product, geographic, operational and sales variables, making it suitable for multidimensional business analysis.



DATA EXPLORATION AND CLEANING

Before loading the dataset into Power BI, I conducted an initial **data exploration and validation process** to understand the structure, fields, data types, and analytical suitability of the dataset.

The first step was to review the **existing data types assigned to each field** and determine whether they were appropriate for analysis. Where necessary, data types and categorizations were adjusted, for an example:

- **Dates/Years** were configured appropriately for time-based analysis.
- **Numerical fields** such as price, mileage, engine size and sales volume were validated as numerical data.
- **Categorical fields** such as model, region, fuel type, transmission, color and classification were treated as dimensions.
- **Geographic fields** were reviewed and assigned appropriate geographic/data categories where applicable.
- Fields were checked to ensure they could be used correctly within **Power BI visuals, filters, aggregations and calculations**.

This preparation helped ensure that the dashboard visuals and DAX calculations were based on **consistent and correctly interpreted data**.



WHAT'S INSIDE THE DASHBOARD

The dashboard is structured around two major analytical experiences:

Executive Summary

The **Executive Summary** provides a high-level view of overall BMW sales performance. It includes:

- **KPI cards** for revenue and units sold
- **Sales and revenue trends over time**
- **Fuel-type distribution**
- **Transmission analysis**
- **Global/regional sales map**
- **Model-wise revenue comparison**
- **Sales classification analysis**

This page is designed for users who want to understand the **overall business picture within seconds**.

DETAILED MODEL ANALYSIS

The **Per-Model Analysis page** provides a deeper view of an individual BMW model.

Users can select a model and investigate:

- **Total units sold**
- **Total revenue**
- **Average price**

- Average mileage
- Average engine size
- Regional performance
- Sales classification
- Fuel-type distribution
 - Transmission split
 - Revenue trends
 - Unit-sales trends
 - Model imagery and visual segmentation

The drill-through/filter-based design allows users to move from “What is happening?” to “Why is it happening?”



TECHNOLOGY STACK

Technology	Purpose
Power BI Desktop	Dashboard development, visualization & interactive analysis
MySQL	Data storage and query environment
DAX	KPI calculations, measures & analytical logic
Git / GitHub	Version control & project documentation
Markdown	README and project documentation



DAX Formulas & New Measures

To support the dashboard's KPIs and analytical visuals, I created **DAX measures** rather than relying only on raw dataset columns.

Key measures included:

- Total Revenue
- Total Units Sold
- Average Price

- **Average Mileage**
- **Average Engine Size**

These measures enabled consistent calculations across different **models, regions, years, fuel types, and other filters**. The use of DAX also made the dashboard more dynamic, allowing KPIs and visualizations to **respond automatically to user selections and drill-downs**.



VISUALS UTILIZED TO ANALYZE THE DATA

A combination of visualizations was used to turn the dataset into an intuitive analytical experience.

 **KPI Cards** : Used to highlight the most important performance indicators, including:

- **Total Revenue**
- **Total Units Sold**
- **Average Price**
- **Average Mileage**
- **Best Performing region**
- **Best Performing Model**
- **Most Sold Vehicle type**
- **Model Name**

 **Geographic Map** : Used to provide a **global perspective of BMW sales performance across regions** by sales volume.

 **Trend Charts** : Used Line chart to analyze how **revenue and unit sales changed over time**, helping identify peaks, declines and model-specific trends.

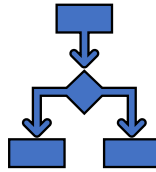
 **Bar Charts** : Used Line and Clustered column chart as a classification view for **Year-wise revenue and sales comparisons**, analyzed through a stacked bar chart to interpret total revenue by model and sales classification.

 **Donut Charts** : Used to understand the distribution of sales by:

- **Fuel Types**
- **Transmission Types**

 **Model Images & Gallery** : Integrated to create a more engaging **model-level analytical experience** and provide visual context for the selected BMW vehicle.

 **Interactive Filters & Drill-Through** : Used to allow users to move seamlessly from the **overall BMW sales picture to individual model analysis**.



STEPS I FOLLOWED TO BUILD THE DASHBOARD

The dashboard was developed through a structured BI workflow:

1 Data Understanding

Reviewed the dataset structure, fields, available dimensions, and measures.

2 Data-Type Validation

Checked and corrected data types, including **numerical, categorical, date, and geographic classifications**.

3 Data Preparation

Ensured fields were appropriately structured for aggregation, filtering, visualization, and analysis.

4 Data Modeling

Loaded the prepared dataset into **Power BI** and established the analytical structure required for the dashboard.

5 DAX Development

Created reusable measures for **revenue, units sold, average price, mileage, and engine size**.

6 KPI Development

Designed executive-level KPI cards to communicate the most important business metrics.

7 Visualization Development

Built trend charts, maps, bar charts, donut charts, and model-level analytical visuals.

8 Interactive Analysis

Added **filters, model selection, drill-through functionality and interactive navigation**.

9 Model-Level Storytelling

Created a detailed page where users can select a BMW model and investigate its performance across different dimensions.

10 Dashboard Refinement

Focused on layout, readability, visual hierarchy, consistency, and ensuring that the dashboard could be understood **quickly by both technical and non-technical users**.



💡 KEY INSIGHTS

The analysis revealed several notable patterns across regions, models, fuel types, and time periods.

❖ The overall analysis indicates strong BMW sales performance, with **total revenue of \$19.01 trillion** generated from approximately **253 million units sold**. **Hybrid vehicles emerged as the most popular vehicle type**, highlighting a strong preference toward more fuel-efficient and alternative-powertrain vehicles within the dataset

🌍 **Regional Performance** : Asia emerged as the strongest overall region, generating approximately \$3.25 trillion in revenue. However, regional leadership varies considerably by model. This indicates that BMW model performance is not uniform across geographic markets.

🚗 **Model Performance** : The 7 Series emerged as the top model by sales volume, with \$23.79 million units sold and Avg price was \$76k, avg milage is 100,792 km with best performing region of “North America “ with \$314.78 Billion. Several other models including Series 3, Series 5, Series 7, i3, i8, M3, M5, and X1 - showed substantial sales activity but differed in their strongest-performing regions.

⚡ **Fuel Preferences** : Hybrid vehicles represented approximately 25.5% of sales, making Hybrid the most prominent fuel category in the dataset. However, fuel-type preferences varied by model. For example, X1 showed strong Electric representation, while several other models were led by Hybrid, Petrol, or Diesel.

⚙️ **Transmission** : Across several models, the dashboard showed a roughly balanced transmission split, suggesting that neither transmission type overwhelmingly dominated the dataset.

📅 **Time Trends** : 2022 represented the peak year overall, with approximately 18 million units sold and \$1.3 trillion in revenue. At the model level, peak years varied, demonstrating that individual BMW models followed different sales cycles and market trajectories.

🎨 **Color Preferences** : Color preferences also varied between models - for example, Silver was prominent for Series 3, while White led for Series 5, and Black was prominent for Series These differences demonstrate how customer preferences can vary across product lines.



◆ THE STORYTELLING

 The dashboard follows a top-down analytical storytelling approach.

It begins with the question:

“How is BMW performing overall?”

The Executive Summary answers this through revenue, units sold, regional performance, fuel mix and time trends.

The analysis then moves toward:

“Which models are driving this performance?”

Model-level comparisons reveal differences between BMW's product lines.

Finally, the dashboard asks:

“Why might individual models be performing differently?”

The detailed analysis explores region, fuel type, transmission, color, price, mileage, sales classification, and historical trends.

This structure allows the viewer to move naturally from overview → comparison → investigation → insight rather than simply viewing disconnected charts.



⚠ CHALLENGES & HOW THEY WERE ADDRESSED

 **Ensuring Correct Data Interpretation:** One of the initial challenges was ensuring that each field had the **appropriate data type and classification** before visualization.

Solution:

I reviewed the dataset before building visuals and adjusted fields to appropriate **numerical, categorical, date, and geographic classifications**.

 **Balancing Multiple Dimensions** : The dataset contains several dimensions, including models, regions, fuel types, transmissions, colors and time. Displaying everything simultaneously could create **visual clutter and make the dashboard difficult to interpret**.

Solution:

The dashboard was divided into an **Executive Summary and Detailed Model Analysis**, allowing users to progressively explore the information.

 **Dynamic Calculations** : Static columns would not provide the flexibility required for interactive analysis.

Solution:

DAX measures were created for important KPIs so that calculations could **dynamically respond to filters and model selections**.

 **Moving from Overview to Detail** : A major analytical requirement was enabling users to investigate individual models without losing the overall context.

Solution:

Interactive **model filtering and drill-through analysis** were incorporated to connect the executive view with detailed model-level insights.



WHY THIS PROJECT MATTERS

This project demonstrates how **Business Intelligence can transform raw transactional data into a decision-support tool**. A dataset containing thousands of individual sales records can be difficult to interpret manually. A well-designed dashboard makes the same information accessible through **KPIs, trends, comparisons, geographic analysis, and interactive exploration**.

From a business perspective, this type of analysis can help stakeholders identify:

- **High-performing markets**
- **Strong and weak product segments**
- **Changing fuel preferences**
- **Model-specific sales patterns**
- **Regional opportunities**
- **Historical sales trends & Potential areas requiring deeper investigation**



STRATEGIC RECOMMENDATIONS

Based on the dashboard analysis, several strategic directions can be considered:

1. Strengthen High-Performing Markets

With **Asia emerging as the strongest overall region**, BMW could further investigate the factors driving performance in this market and identify opportunities to strengthen successful models and customer segments.

2. Focus on High-Performing Models

The strong sales performance of the **7 Series** indicates an opportunity to study the characteristics behind its success and identify whether similar strategies could benefit other models.

3. Monitor the Shift Toward Electrification

With **Hybrid representing approximately 25.5% of sales** and Electric leading for some individual models, BMW should continue monitoring changes in **powertrain preferences** across markets.

4. Develop Market-Specific Strategies

Since the strongest region varies by model, a single global strategy may not be optimal. BMW could consider **model-specific regional strategies** based on local demand patterns.

5. Monitor Sales Cycles

The variation in peak years across models suggests that product performance changes over time. Tracking these trends can help with **product lifecycle planning, marketing campaigns, inventory allocation, and future model strategy**.

6. Use Customer Preference Patterns

Differences in **color, fuel type, transmission, price, and mileage patterns** can provide additional signals for understanding customer preferences and supporting more targeted product and marketing decisions.

7. Continue Deep-Dive Analysis

The dashboard should be treated as a **decision-support starting point rather than an endpoint**. Further analysis could investigate profitability, customer segments, pricing elasticity, market growth, and relationships between vehicle characteristics and sales performance.